



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 19 summarizes the current fairness state for upload 19. Accuracy changed by +0.954 and fairness score changed by +0.942. Strong fairness improvement with no measurable accuracy loss. The best balanced model is Pipeline_mitigated_reweighting with combined score 0.947.

ACCURACY	FAIRNESS SCORE	COMBINED SCORE	DISPARATE IMPACT
0.954 ▲ +0.954	0.942 ▲ +0.942	0.947 Pipeline_mitigated_reweighting	0.754 Ratio (DfR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	german_rf_baseline.pkl	comparison

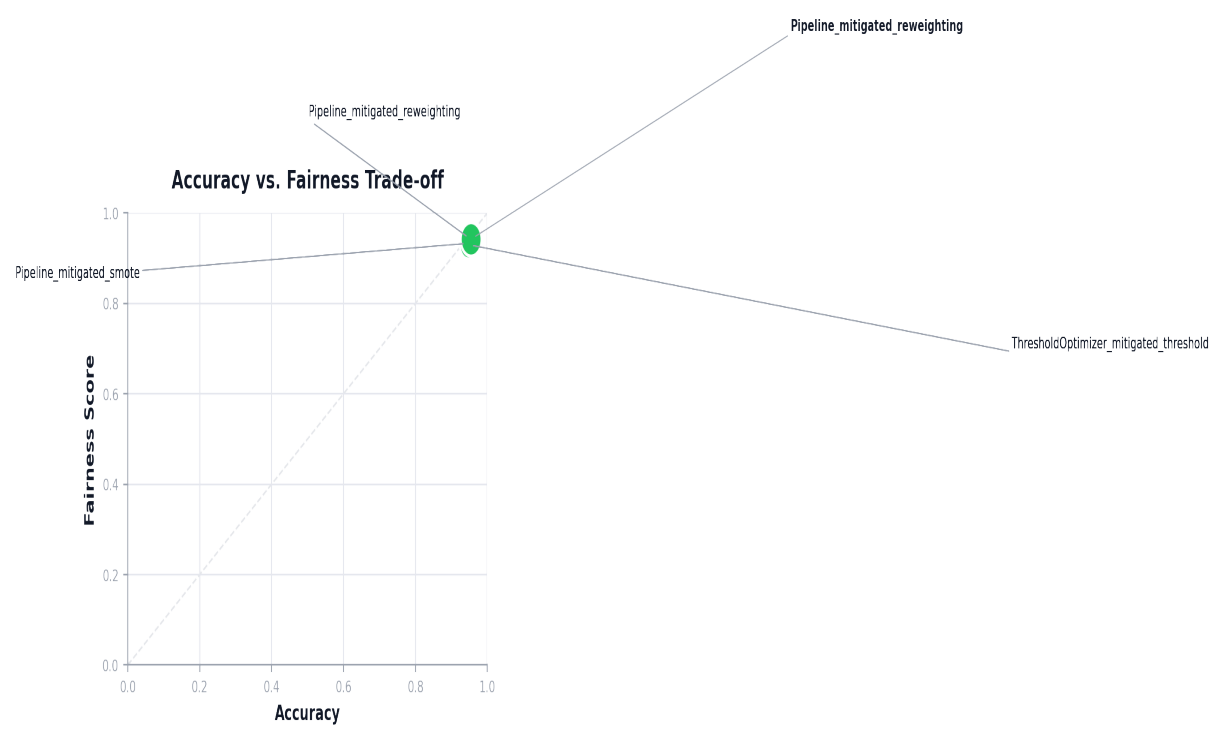
Fairness Metrics

Metric	Value	Interpretation
Demographic Parity Difference (DPD)	0.084	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.033	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.754	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

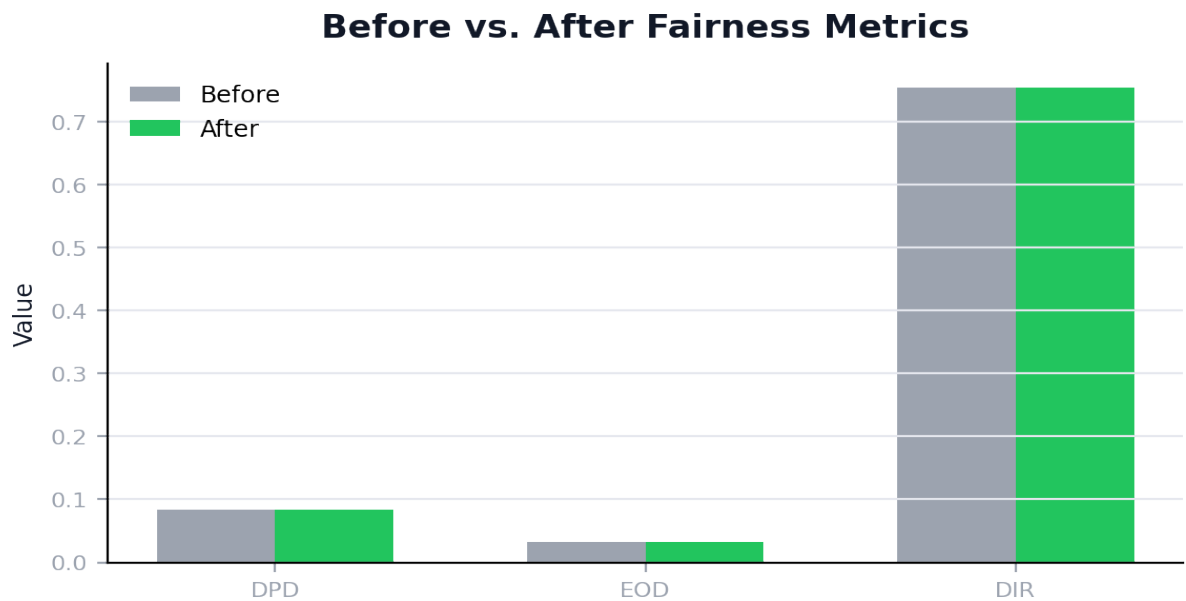
Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ Pipeline_mitigated_reweighting	0.954	0.942	0.084	0.033	0.754	0.947
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★ Pipeline_mitigated_reweighting	0.954	0.942	0.084	0.033	0.754	0.947
Pipeline_mitigated_smote	0.947	0.933	0.088	0.046	0.753	0.939
Pipeline_mitigated_smote	0.947	0.933	0.088	0.046	0.753	0.939
Pipeline_mitigated_smote	0.947	0.933	0.088	0.046	0.753	0.939
ThresholdOptimizer_mitigated_threshold	0.947	0.930	0.092	0.048	0.736	0.937
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ThresholdOptimizer_mitigated_threshold	0.947	0.930	0.092	0.048	0.736	0.937

Accuracy vs. Fairness Trade-off



Before vs. After Fairness Metrics



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.